INTERIM FEDERAL SPECIFICATION

TRAP, DRAIN, PLUMBING FIXTURE (P-TRAP)
(LAND USE)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

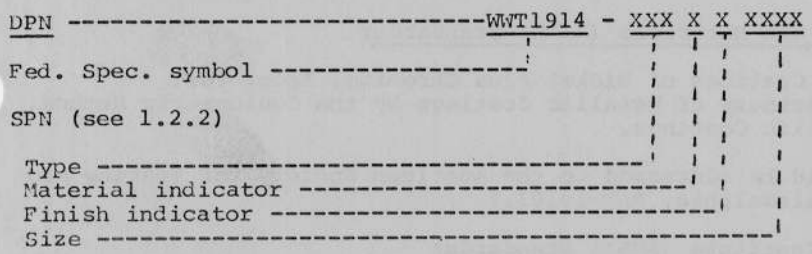
1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for general use P-traps.

1.1.1 Specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available. This specification is intended to cover P-traps generally used by the Federal Government.

1.2 Classification.

1.2.1 Definitive part number (DPN). Items furnished in accordance with this specification shall be identified by a DPN. This number is intended for cataloging and ordering purposes; it is not intended for surface printing on items in accordance with this specification. The DPN shall consist of the Federal Specification symbol plus the specification part number (SPN) (see 1.2.2). The DPN shall be written as indicated below (see 6.2 and 6.3).



1.2.2 SPN. The SPN shall be a 9 position designator which identified items covered by this specification. The designator shall identify the type, material, finish, and size, respectively in accordance with table I.

TABLE I. SPN identification

Type/SPN	Material/indicator	Finish/indicator	Size (inch)	Reference paragraph
106	Copper alloy/B	Chrome/C	1-1/2	3.11.1
114	Copper alloy/B	Chrome/C	1-1/2	3.11.2
115	Copper alloy/B	Chrome/C	1-1/4	3.11.3

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Setup.

FSC 4510

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:24Z
Check the source to verify that this is the current version before use.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

B 456 - Electrodeposited Coatings of Nickel Plus Chromium, Spec. for.
B 504 - Measuring the Thickness of Metallic Coatings by the Coulometric Method.
B 571 - Adhesion of Metallic Coatings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American National Standards Institute (ANSI) Standards:

B 16.23 - Cast-Bronze Solder-Joint Drainage Fittings.
B 16.29 - Wrought-Copper and Wrought-Copper Alloy Solder-Joint Drainage Fittings.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:24Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General.

3.1.1 Cast copper alloy P-traps. Except as otherwise specified herein, cast copper alloy P-traps shall conform to the requirements of ANSI B 16.23.

3.1.2 Wrought copper alloy P-traps. The thickness of wrought copper alloy P-traps shall be 0.032 + 0.003 inches prior to bending. Except for thickness, traps shall conform to ANSI B 16.29 for material and end dimensions and to ANSI B 16.23 for other dimensions.

3.2 First article. After award of contract, the supplier shall furnish, within the time limit specified in the contract, one of each type P-trap he proposes to supply, for first article inspection. The sample shall be supplied to the contracting officer or his authorized representative for approval in accordance with 4.2. Approval of the first article sample authorizes the commencement of production but does not relieve the supplier of his obligation to meet the requirements of the specification for the remaining units delivered under the contract. Upon completion of first article inspection and acceptance, the approved sample shall be maintained in a location available to both the authorized representative of the contracting officer and the supplier for the duration of the contract period.

3.2.1 Certification. Following acceptance of the first article, and in lieu of the tests specified in 4.5, the supplier or his authorized representative may certify to the contracting officer or his authorized representative, that all P-traps and component parts in each lot are identical and produced under the same conditions as the first article.

3.3 Material. Material not definitely specified shall be of the quality normally used for the intended purpose in commercial practice.

3.4 Water seal depth. Depth of water seal shall be not less than 2 inches.

3.5 Inlet connection. The inlet connection shall be slip joint (SJ). The inlet connection shall have an elastomeric washer and a copper alloy or die-cast zinc nut. The nut shall have flat sides for wrench grip. Size of nut shall be compalible with the trap size.

3.6 Cleanout plug. Traps shall be equipped with a copper alloy cleanout plug. The plug shall have flats for wrench grips. The plug shall be furnished with a washer to seal the joint. The cleanout shall not form a pocket in the trap or obstruct the flow.

3.7 Finish.

3.7.1 Internal surfaces. Internal surfaces shall have an as-cast surface for cast parts and a smooth mill finish for wrought parts.

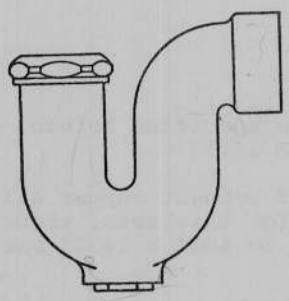
3.7.2 Exterior surfaces. Exposed exterior surfaces of traps, plugs, and other hardware shall be bright chromium plated in accordance with SC1 of ASTM B 456.

3.8 Torque performance. Trap joints with gaskets shall withstand a torque of 10 foot-pounds and trap joints without gaskets shall withstand a torque of 40 foot-pounds without failure when tested in accordance with 4.5.3.

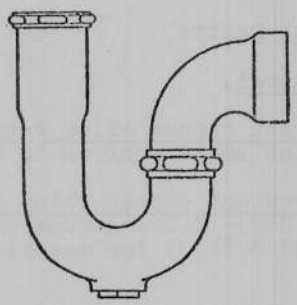
3.9 Passage through traps. P-traps shall be capable of passing a 1 inch diameter ball when tested in accordance with 4.5.4.

3.10 In addition to the requirements specified in 3.1 thru 3.9, P-traps shall conform to the applicable specific requirements of figure 1 and 3.11.

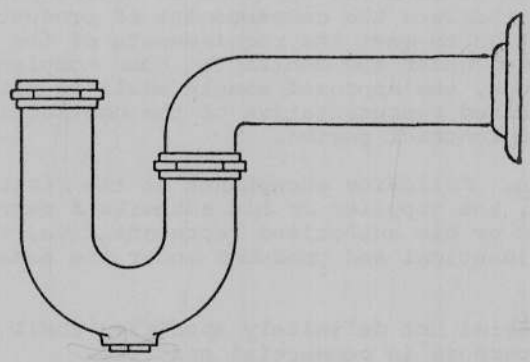
Source: https://assist.dla.mil -- Downloaded: 20260-26T12:24Z
Check the source to verify that this is the current version before use.



Type 106



Type 114



Type 115

FIGURE 1

3.11 Specific requirements.

3.11.1 Type 106 (SPN 106). One piece cast copper alloy P-trap. Slip joint inlet (SJ) to female threaded outlet (F) connection. Size 1-1/2 inch outside diameter (OD).

3.11.2 Type 114 (SPN 114). Two piece metal to metal swivel joint. Wrought metal P-trap with cast elbow. SJ inlet to F outlet. Seven and one-half (7-1/2) inches overall length. Size 1-1/2 inch OD.

3.11.3 Type 115 (SPN 115). Two piece wrought copper alloy with metal to metal swivel joint. Length of elbow extension tube shall be not less than 11 inches measured from the centerline of the traps inlet to the tube outlet. Tube furnished with escutcheon. Trap inlet connection shall be SJ; tube connection to waste pipe shall be straight tube for use with adapter. Trap and tube size 1-1/4 inch OD.

3.12 Workmanship. All parts shall be free of cracks, dents, sharp edges, and welding flux or scale. Plating shall be uniform and without visible void areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:24Z
Check the source to verify that this is the current version before use.

4.2 First article sample inspection. The first article shall be examined in accordance with 4.4 and tested as specified in 4.5. Each lot shall be examined in accordance with 4.4 (see 3.2.1).

4.3 Lot. A lot shall consist of all P-traps of the same DPN offered for delivery at one time.

4.3.1 Sampling for examination. Select sample size in accordance with level S-1 in MIL-STD-105.

4.4 Examination. Examine traps and parts for defects listed in table II. The acceptable quality levels (AQL's) for defects shall be 2.5 percent defective for major defects and 6.5 percent defective for total defects.

TABLE II. Classification of defects, end item

Defects	Major	Minor
DPN not as specified	X	
Components missing	X	
Cracks in traps or components	X	
Dimensions not as specified	X	
Water seal not as specified	X	
Chrome plating not uniform		X
Damage or defects affecting function or serviceability	X	
Damage or defects not affecting function or serviceability		X

4.5 Test methods.

4.5.1 Chromium plate thickness. Test chromium plate thickness in accordance with ASTM B 504.

4.5.2 Chromium plate adhesion. Test adhesion of chromium plate in accordance with ASTM B 571.

4.5.3 Trap joints. Using a suitable torque wrench apply a torque of 10 foot-pounds at the joints of traps with gasket, and 40 foot-pounds for trap joints without gaskets (see 3.8).

4.5.4 Drain test. With water flowing, through the trap, insert a 1 inch diameter solid ball at the inlet end of the P-trap (see 3.9).

4.6 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping conatiner fully prepared for delivery, selected at random, just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2, with an AQL of 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Any component missing, damaged or not conforming to the applicable specification or requirements stated herein.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.
Contents	Quantity per container is more or less than required. Gross weight exceeds requirements of the box specification.

Source: https://assist.dla.mil -- Downloaded: 2026-04-26T12:24Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Each drain trap shall be packaged in a close-fitting box conforming to PPP-B-636, class weather resistant, style optional. Close and waterproof seal the boxes in accordance with the appendix to PPP-B-636. Alternatively, each drain trap shall be packaged in a closefitting box conforming to PPP-B-566, PPP-B-665 or to PPP-B-676, and further packaged in a sealed bag constructed from film conforming to L-P-378, a minimum of 1.5 mils thick.

5.1.2 Level B. Each drain trap shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, or to PPP-B-676. Closure of the boxes shall be in accordance with the applicable box specification or appendix thereto.

5.1.3 Level C. Drain traps shall be packaged to insure protection during shipment and safe delivery to destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twenty-five drain traps of like description, packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class weather resistant. Close and waterproof seal the boxes in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Twenty-five drain traps of like description, packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. Closure of the boxes shall be in accordance with Method II of the appendix to PPP-B-636.

5.2.3 Level C. Drain traps of like description packaged as specified in 5.1 shall be packed to comply with the National Motor Freight rules or the Uniform Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. P-traps in accordance with this specification are intended for interior use in general purpose waste systems.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Definitive part number (see 1.2.1).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) When special markings are required (see 5.3).

6.3 Example of DPN.



Source: https://assist.dla.mil -- Downloaded: 202640-26T12:24Z
Check the source to verify that this is the current version before use.